

COMPUTER NETWORKS

PROJECT REPORT

SUBMITTED TO: -

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TITLE:

Home Network Using Wireless Router in Cisco Packet Tracer

INTRODUCTION:

A computer network connects different devices so they can communicate and share data. In this lab, a small network was created using a wireless router, PCs, cloud, cable modem, laptop, switch , coaxial cable and server in Cisco Packet Tracer.

OBJECTIVE:

To create a small network using a wireless router, PCs, cloud ,laptop, switch and server and test communication between devices.

COMPONENTS USED:

- 1 Wireless Router (WRT300N)

- 2 PCs
- 1 Cloud
- 1 Cable Modem
- 1 Server
- Copper Straight Through Cables
- 1 laptop □ 2 Switches
- Coaxial Cable

DESCRIPTION OF DEVICES:

Router: Connects devices and manages communication between networks.

PCs: End-user devices used for sending and receiving data.

Cloud: Represents the internet or ISP network.

Cable Modem: Connects the router to the internet service provider.

Server: Provides services and stores data for the network.

Laptop: Portable device for network access and testing.

Switches: Connect multiple devices in a LAN and forward data to the correct device.

Coaxial Cable: Carries internet data between the modem and ISP network.

NETWORK TOPOLOGY:

The PCs and Laptop are connected to the network through two Switches. One Switch connects to the Wireless Router, which manages routing and wireless communication. The Router is connected to the Cloud through a Cable Modem for internet access. A Server is connected separately to provide services. IP addresses are assigned to all devices for communication.

IP CONFIGURATION:

Device	IP Address	Subnet Mask	Default Gateway

PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1
Router	192.168.1.1	255.255.255.0	—
Laptop	192.168.1.4	255.255.255.0	192.168.1.1
Server	192.168.1.100	255.255.255.0	192.168.1.1

PROCEDURE:

1. Added all devices in Cisco Packet Tracer.
2. Connected devices using cables.
3. Configured IP addresses.
4. Set subnet mask and default gateway.
5. Tested connectivity using Simple PDU.
6. Verified successful communication between PCs.

WORKING OF THE NETWORK:

When a packet is sent from a PC or Laptop, it travels through the connected Switch to the Wireless Router. The Router forwards the packet to the destination device within the LAN, or to the Server if requested. For internet-bound traffic, the Router sends the packet through the Cable Modem and Cloud to the ISP network. The Switches manage data flow within their LAN segments by forwarding packets only to the intended device using MAC addresses, while the Router handles communication between different networks and devices.

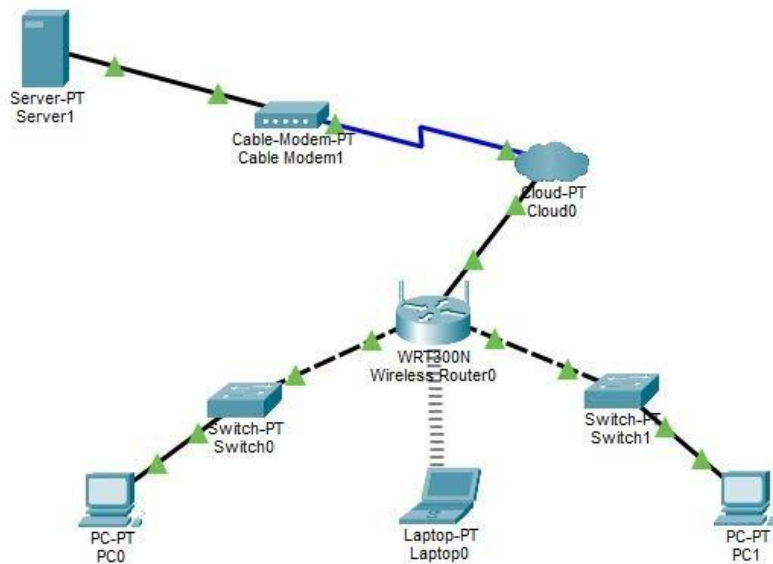
ADVANTAGES:

- ✓ Easy communication between devices
- ✓ Fast data sharing
- ✓ Internet connectivity

- ✓ Centralized network management

RESULT:

“The network was successfully configured and devices communicated successfully.”



PING PC0:

```

Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

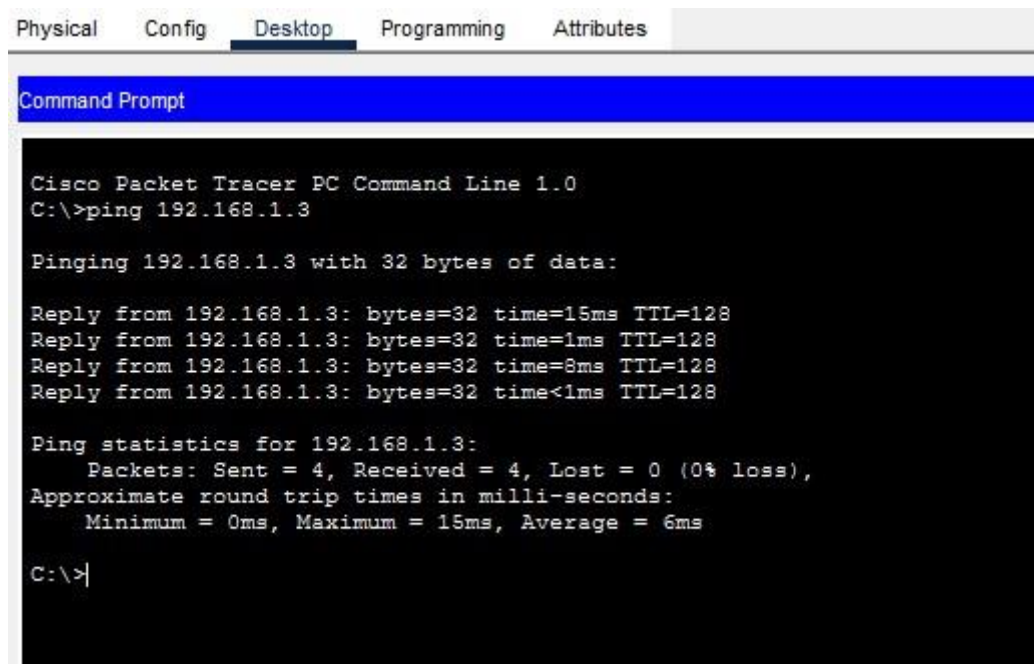
Reply from 192.168.1.2: bytes=32 time=6ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time=11ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 4ms

C:\>

```

PING PC1:



The screenshot shows the Cisco Packet Tracer interface with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

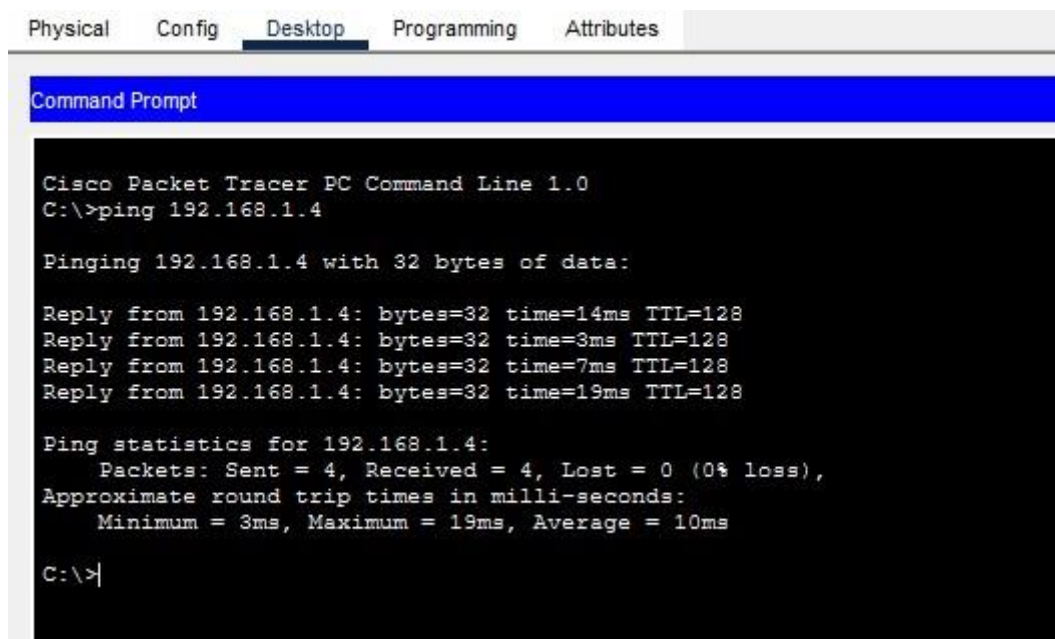
Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=15ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=8ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 6ms

C:\>
```

PING LAPTOP:



The screenshot shows the Cisco Packet Tracer interface with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=14ms TTL=128
Reply from 192.168.1.4: bytes=32 time=3ms TTL=128
Reply from 192.168.1.4: bytes=32 time=7ms TTL=128
Reply from 192.168.1.4: bytes=32 time=19ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 19ms, Average = 10ms

C:\>
```

PING SERVER:

```
C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 192.168.1.100: bytes=32 time=6ms TTL=128
Reply from 192.168.1.100: bytes=32 time=5ms TTL=128
Reply from 192.168.1.100: bytes=32 time=7ms TTL=128
Reply from 192.168.1.100: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 7ms, Average = 6ms

C:\>|
```

CONCLUSION:

This lab helped in understanding basic networking concepts, IP addressing, routing, and packet communication using Cisco Packet Tracer.